

Notes: Kleven, Landais, and Søgaaard (AEJ Applied, 2019)

Children and Gender Inequality: Evidence from Denmark

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Administrative data	3
2.2	Event-study sample	4
2.3	Labor-market outcomes	4
2.4	Child penalty	5
3	Empirical specifications (all equations + interpretation)	5
3.1	Baseline event-study regression	5
3.2	Percentage event effects and child penalties	6
3.3	Dynamic decomposition of gender inequality	6
3.4	Intergenerational transmission specification	6
4	Identification and mechanisms (what makes the estimates credible)	7
4.1	Event-study logic	7
4.2	Why men are the main comparison group	7
4.3	Identification checks	7
4.4	Mechanism: family-friendly work choices	8
4.5	Limitations	8
5	Main results	8
5.1	Impacts of children	8
5.2	Anatomy of child impacts	9
5.3	Decomposition of gender inequality	9
5.4	Education and pre-child effects	9

5.5	Intergenerational transmission	9
6	Figures and tables (visuals + reading guide)	10
6.1	Figure 1: Impacts of Children and Figure 2: Impacts of Children in the Very Long Run	10
6.2	Figure 3: Anatomy of Child Impacts and Figure 4: Earnings Penalties by Birth Cohort and Extrapolation	11
6.3	Figure 5: Decomposing Gender Inequality in Earnings and Figure 6: Intergenerational Transmission of Child Penalties without Controls	12
6.4	Figure 7: Intergenerational Transmission of Child Penalties with Controls	13
7	Short summary	13
8	Conclusion	13

1 Big picture

- **Question.** How much of remaining gender inequality in the labor market is caused by children? The paper studies whether the arrival of children changes women’s labor-market paths relative to men’s, and whether those changes can explain the persistent gender earnings gap.
- **Empirical setting.** Denmark is an unusually clean setting because the authors use full-population administrative data from 1980–2013, with links across individuals, children, spouses, firms, education, occupation, earnings, and hours worked. For the intergenerational analysis, the data extend back to 1964.
- **Core empirical strategy.** The main design is an event study around the birth of the first child. The authors compare male and female outcomes before and after parenthood, controlling flexibly for age and calendar-year effects.
- **Main result.** Men and women evolve similarly before the first child. Immediately after childbirth, women experience a sharp drop in earnings, hours, participation, and wages, while men are almost unaffected. The long-run earnings child penalty is about **20 percent**.
- **Mechanisms.** The earnings penalty operates through three broad labor-market margins: hours worked, labor force participation, and wage rates. It is also reflected in job characteristics: after children, women fall behind in occupational rank, become less likely to be managers, and move toward more family-friendly sectors and firms.
- **Dynamic decomposition.** The authors decompose aggregate gender inequality into child-related and residual components. The share of earnings inequality explained by child penalties rises from about **40 percent in 1980** to about **80 percent in 2013**. The striking implication is that today’s remaining gender inequality is, to a first approximation, about children.
- **Intergenerational result.** Female child penalties are strongly related to the relative labor supply of maternal grandparents, but not paternal grandparents. Women who grew up in more traditional families, where the grandmother worked relatively little compared with the grandfather, incur larger child penalties when they become mothers.
- **Economic interpretation.** Parenthood changes the allocation of market work and child-care responsibilities in a strongly gendered way. Even in a country with extensive family policies and high female labor force participation, children create persistent inequality between men and women.
- **Takeaway.** The paper shifts the gender-gap question away from only “equal pay for equal work” toward a more fundamental issue: after children, equal work often stops being feasible because family responsibilities are unequally allocated.

2 Data and measurement (key objects)

2.1 Administrative data

- **Main period.** The core analysis uses Danish administrative data for the full population from 1980 to 2013.

- **Intergenerational period.** For transmission across generations, the authors use additional hours-worked information going back to 1964.
- **Linked records.** The data link individuals to spouses, children, parents, grandparents, firms, occupations, education, earnings, and labor supply.
- **Why Denmark matters.** Denmark has generous family policies, subsidized child care, job-protected parental leave, and high female labor force participation. Therefore, large child penalties in Denmark are especially informative: they arise even in an institutionally supportive environment.

Interpretation. The data make it possible to follow the same people before and after childbirth and to connect their labor-market outcomes to family structure and childhood environment. This is what allows the paper to move beyond cross-sectional gender-gap decompositions.

2.2 Event-study sample

- **Event.** The event is the birth of the first child. Event time is denoted by t , with $t = 0$ in the year of first childbirth.
- **Baseline window.** The main balanced panel observes parents from five years before to ten years after the first child, so $t = -5, \dots, 10$.
- **Birth cohorts.** The core sample focuses on first births between 1985 and 2003.
- **Sample size.** The baseline sample contains roughly 470,000 births and about 15 million individual-year observations.
- **Relationship status.** The authors do not restrict the sample to married or cohabiting parents. Individuals are followed whether they are married, cohabiting, separated, divorced, or not in a couple.

Interpretation. The design is built to estimate what happens to labor-market trajectories when individuals become parents for the first time. The key comparison is not women with children versus women without children in the cross-section, but women’s own trajectories before and after the child, compared with men’s trajectories.

2.3 Labor-market outcomes

- **Earnings.** Gross labor earnings before taxes and transfers.
- **Labor force participation.** Whether the person participates in the labor market.
- **Hours worked.** Administrative hours are measured using contributions to the ATP pension scheme, which has been mandatory since 1964 and is reported by employers.
- **Wage rate.** Earnings divided by the administrative measure of hours worked, conditional on participation.
- **Occupation and management.** Occupational rank is coded in five levels, from unskilled labor to top manager. The authors also study the probability of being a manager.

- **Sector and firm family friendliness.** The public sector is used as one family-friendly work environment. The authors also proxy firm family friendliness using whether management includes women with young children.

Measurement issue. The ATP hours measure is capped and discrete, so it may understate marginal hours changes among people continuously working at the highest hours category. However, it captures large labor-supply adjustments such as part-time work and work interruptions, which are central for the child-penalty question.

2.4 Child penalty

The paper defines a child penalty as the amount by which women fall behind men because of children. For outcome Y , the child penalty at event time t is conceptually:

$$P_t = \frac{\hat{\alpha}_t^m - \hat{\alpha}_t^w}{\mathbb{E}[\tilde{Y}_{ist}^w | t]},$$

where $\hat{\alpha}_t^m$ and $\hat{\alpha}_t^w$ are the estimated event-time effects for men and women, and $\tilde{Y}_{ist}^w$ is women's predicted counterfactual outcome absent the event-time effect.

Read. P_t is not simply the raw gender gap at event time t . It is the event-study estimate of how much women fall behind men because of the child, scaled by women's counterfactual outcome.

3 Empirical specifications (all equations + interpretation)

3.1 Baseline event-study regression

For individual i , gender $g \in \{m, w\}$, calendar year s , and event time t , the baseline event-study specification is

$$Y_{ist}^g = \sum_{j \neq -1} \alpha_j^g \mathbf{1}[j = t] + \sum_k \beta_k^g \mathbf{1}[k = \text{age}_{is}] + \sum_y \gamma_y^g \mathbf{1}[y = s] + \nu_{ist}^g. \quad (1)$$

- Y_{ist}^g is the labor-market outcome.
- α_j^g are event-time coefficients.
- The omitted event year is $t = -1$, the year before the first child.
- Age fixed effects absorb life-cycle patterns.
- Year fixed effects absorb aggregate time trends, wage inflation, and business-cycle conditions.
- The regression is estimated separately for men and women.

Read. The coefficient α_j^g measures how the outcome for gender g at event time j differs from the year before childbirth, after removing age and calendar-year patterns. The identifying variation comes from differences in the age at which people have their first child.

3.2 Percentage event effects and child penalties

Because equation (1) is estimated in levels, the authors convert the level effects into percentage effects:

$$P_t^g = \frac{\hat{\alpha}_t^g}{\mathbb{E}[\tilde{Y}_{ist}^g | t]}.$$

The gendered child penalty is

$$P_t = \frac{\hat{\alpha}_t^m - \hat{\alpha}_t^w}{\mathbb{E}[\tilde{Y}_{ist}^w | t]}. \quad (2)$$

Read. P_t^g gives the event effect for men or women separately. P_t gives the female penalty relative to men. If men are unaffected and women fall by 20 percent, then the child penalty is roughly 20 percent.

Why levels rather than logs. Estimating in levels keeps nonparticipants in the sample. This is crucial because participation itself is one of the margins through which child penalties operate.

3.3 Dynamic decomposition of gender inequality

To decompose aggregate gender inequality over time, the authors extend the event study to allow event-time effects to vary by calendar year:

$$Y_{ist}^g = \sum_y \sum_{j \neq -1} \alpha_{yj}^g \mathbf{1}[j = t] \mathbf{1}[y = s] + \sum_k \beta_k^g X_{kis}^g + \nu_{ist}^g. \quad (3)$$

Let the mean gender gap in earnings in year s be

$$\Delta_s = \frac{\mathbb{E}[Y_{ist}^m | s] - \mathbb{E}[Y_{ist}^w | s]}{\mathbb{E}[Y_{ist}^m | s]}.$$

The decomposition can be written schematically as

$$\hat{\Delta}_s = \underbrace{\frac{\mathbb{E}[P_{st} \tilde{Y}_{ist}^w | s]}{\mathbb{E}[\tilde{Y}_{ist}^m | s]}}_{\text{child-related inequality}} + \underbrace{\text{different returns to non-child covariates}}_{\text{unexplained residual}} + \underbrace{\text{differences in non-child covariates}}_{\text{explained residual}}. \quad (4)$$

Read. The first term is the part of gender inequality caused by child penalties. The remaining terms are residual gender inequality not attributed to the post-child event effect. When the authors add education controls, the residual can be further split into education-related and other residual components.

3.4 Intergenerational transmission specification

For the intergenerational analysis, the authors compress the post-child period into an after-child indicator and estimate how child penalties differ by the relative labor supply of grandparents:

$$\begin{aligned} Y_{ist}^g = & \sum_q \alpha_q^g \mathbf{1}[t \geq 0] \mathbf{1}[\text{grand}_{iq}^m = q] + \sum_k \beta_k^g \mathbf{1}[k = \text{age}_{is}] + \sum_y \gamma_y^g \mathbf{1}[y = s] \\ & + \sum_q \zeta_q^g \mathbf{1}[\text{grand}_{iq}^m = q] + \delta^g X_i^m + \eta^g \mathbf{1}[t \geq 0] X_i^m + \nu_{ist}^g. \end{aligned} \quad (5)$$

- grand_{iq}^m is the quantile of relative labor supply among maternal grandparents.
- A higher quantile means a more modern gender division of labor in the mother's childhood home.
- The same specification is estimated for paternal grandparents by replacing maternal-grandparent variables with paternal-grandparent variables.
- X_i contains controls such as grandparent education, wealth rank, region, and cohort, depending on the specification.

Read. The coefficients α_q^g ask whether women's child penalties differ depending on how traditional or modern their parents' household was one generation earlier.

4 Identification and mechanisms (what makes the estimates credible)

4.1 Event-study logic

- The event study does not require fertility to be random in a literal sense. It requires that, absent childbirth, labor-market outcomes would have evolved smoothly around the event.
- The key visual evidence is the parallel pre-trend: men and women evolve similarly before the birth of the first child, then diverge sharply at childbirth.
- The discontinuity at childbirth is hard to explain with smooth life-cycle trends, because age and calendar year are controlled nonparametrically.
- The event-study design is especially persuasive for short-run effects. Long-run effects require stronger assumptions because outcomes may diverge for other reasons far after the event.

4.2 Why men are the main comparison group

- In the baseline child penalty, men serve as the comparison for women.
- This is useful because men experience the same family event, but their labor-market trajectories barely change around childbirth.
- If men were strongly affected by childbirth, the child penalty would understate or distort the effect on women. The paper finds that men's effects are very small relative to women's.

4.3 Identification checks

- **Difference-in-differences event study.** The paper assigns placebo births to people who never have children, using the observed distribution of ages at first birth. Women with and without children have similar pre-trends, and the long-run female earnings impact is about 20.6 percent, close to the baseline estimate.
- **Sex-mix IV comparison.** The paper compares an event study around the third child to an instrumental-variables estimate using the sex mix of the first two children as an instrument for having a third child. The IV and event-study estimates are closely aligned.

- **Quantile regressions.** The paper checks that mean impacts are not driven only by the top of the earnings distribution. Median effects are broadly similar to mean effects.

Interpretation. These checks support the idea that the event-study approach captures causal labor-market effects of children rather than only selection into parenthood.

4.4 Mechanism: family-friendly work choices

- After childbirth, women move into jobs and firms that appear more compatible with family responsibilities.
- Women become more likely to work in the public sector, which in Denmark is associated with family-friendly conditions such as more flexible hours and leave policies.
- Women also become more likely to work at firms where management includes women with young children, a proxy for a family-friendly firm culture.
- These job choices help explain why the child penalty is not only an hours or participation penalty, but also a wage-rate penalty.

4.5 Limitations

- The event study identifies post-child effects, not anticipation effects before childbirth. If women choose education or career paths in anticipation of future motherhood, the estimated child penalties are lower bounds on lifetime child-related costs.
- The intergenerational results are correlational, although rich controls make a simple omitted-variable explanation less likely.
- The paper is deliberately cautious about welfare. A child penalty may reflect constraints, social norms, discrimination, comparative advantage, or voluntary preferences. The evidence documents the magnitude and mechanisms, but does not by itself determine the normative interpretation.

5 Main results

5.1 Impacts of children

- Women's earnings fall sharply at the birth of the first child and never recover to the male trajectory within the observed horizon.
- Ten years after first childbirth, the earnings child penalty is about 19.4 percent.
- Hours worked, participation, and wage rates all fall for women relative to men. The three margins are roughly comparable in importance.
- Extending the horizon to 20 years shows that the earnings penalty remains very persistent. Hours worked begin to converge somewhat, but wage rates continue to diverge, leaving the overall earnings gap stable.

Interpretation. Parenthood creates a durable gender gap, not a temporary maternity-leave gap. This is the central empirical fact of the paper.

5.2 Anatomy of child impacts

- Women's occupational rank falls relative to men's after childbirth.
- The probability of becoming a manager falls sharply for women relative to men; the long-run manager penalty is about 26 percent.
- Women become more likely to work in the public sector after childbirth.
- Women become more likely to work in firms with female managers who have children.

Interpretation. Children affect not only how much women work, but also where and in what kinds of jobs they work. The wage penalty is consistent with movement toward jobs with more family amenities and lower pecuniary rewards.

5.3 Decomposition of gender inequality

- The authors decompose earnings inequality into child-related and residual components.
- Child-related earnings inequality rises from about 18 percent to almost 20 percent between 1980 and 2013.
- Total gender earnings inequality falls from about 46 percent to about 24 percent over the same period.
- Because non-child inequality falls while child-related inequality remains large, the share of inequality due to children rises from about 40 percent to about 80 percent.

Interpretation. The gender gap has not disappeared because the part of the gap associated with children has not disappeared. Instead, other sources of inequality have shrunk around it.

5.4 Education and pre-child effects

- The paper distinguishes post-child effects from possible pre-child anticipation effects.
- Education-related gender inequality falls dramatically over time, partly because women catch up to and overtake men in education levels.
- Adding education controls barely changes the child-related component, because the child effect is identified from within-person changes around childbirth.

Interpretation. Earlier generations may have paid more of the child cost upfront through education and career choices. More recent generations invest more similarly before children, but still experience large penalties after motherhood.

5.5 Intergenerational transmission

- The female child penalty is smaller for women whose maternal grandparents had a more modern division of labor.
- Moving from the most traditional to the most modern maternal-grandparent quintile reduces the earnings child penalty by roughly six percentage points, about one-third of the total penalty.

- The paternal-grandparent relationship is essentially flat.
- The result survives controls for grandparent education, wealth, region, and cohort.
- Controlling for the parents' own fertility does not remove the pattern, suggesting that transmission operates through family-career preferences conditional on fertility, not simply through having more children.

Interpretation. The results are consistent with an influence of childhood environment on female gender identity. Women who grow up observing a traditional maternal household division of labor are more likely to reproduce traditional child penalties in their own adulthood.

6 Figures and tables (visuals + reading guide)

The paper's main text reports its evidence through figures rather than regression tables. The visuals below keep the title for each figure, omit visible figure numbering in the note, and place adjacent visuals two side-by-side where possible.

6.1 Figure 1: Impacts of Children and Figure 2: Impacts of Children in the Very Long Run

Figure 1: Impacts of Children

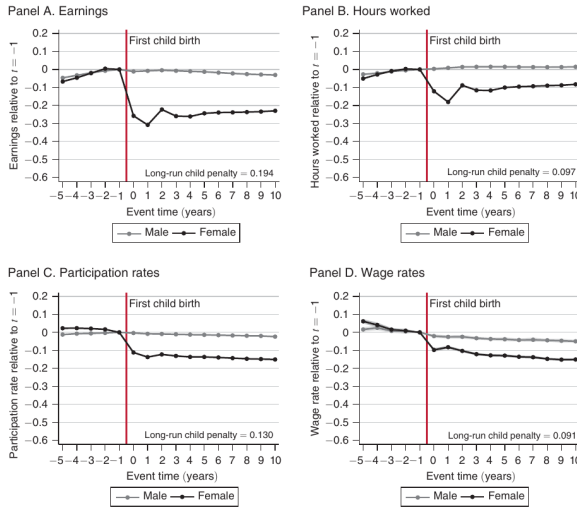
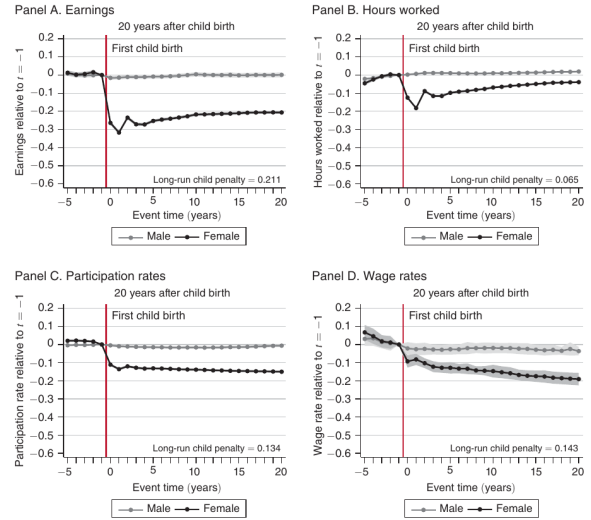


Figure 2: Impacts of Children in the Very Long Run



Read. The first visual is the paper's central event-study evidence. Women and men are on similar pre-child paths, but women's earnings, hours, participation, and wage rates fall sharply after first childbirth. The very-long-run visual shows that the earnings penalty persists for 20 years: hours recover somewhat, but wage rates continue to diverge.

6.2 Figure 3: Anatomy of Child Impacts and Figure 4: Earnings Penalties by Birth Cohort and Extrapolation

Figure 3: Anatomy of Child Impacts

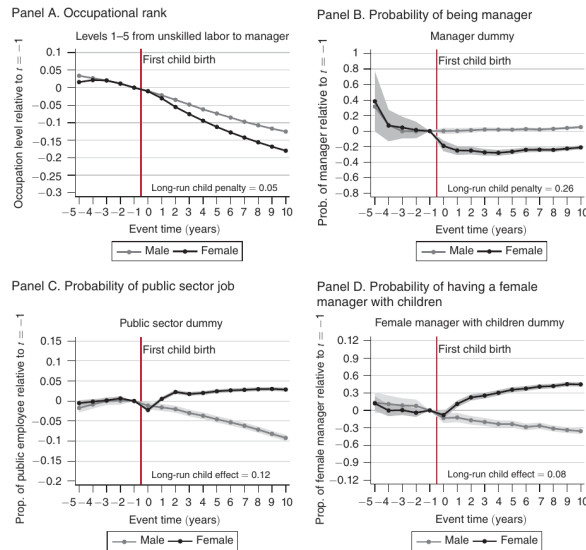
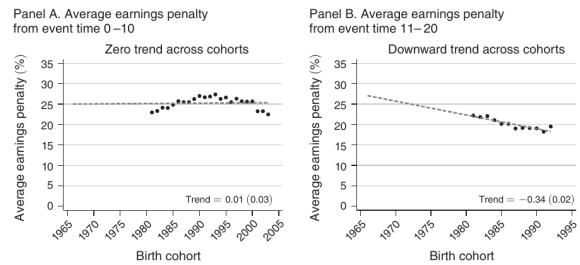


Figure 4: Earnings Penalties by Birth Cohort and Extrapolation



Read. The anatomy visual shows that child penalties operate through job characteristics as well as hours: women fall behind in occupational rank and management probability, and move toward public-sector and family-friendly firms. The birth-cohort visual explains how the authors extrapolate penalties for older cohorts in the dynamic decomposition.

6.3 Figure 5: Decomposing Gender Inequality in Earnings and Figure 6: Inter-generational Transmission of Child Penalties without Controls

Figure 5: Decomposing Gender Inequality in Earnings

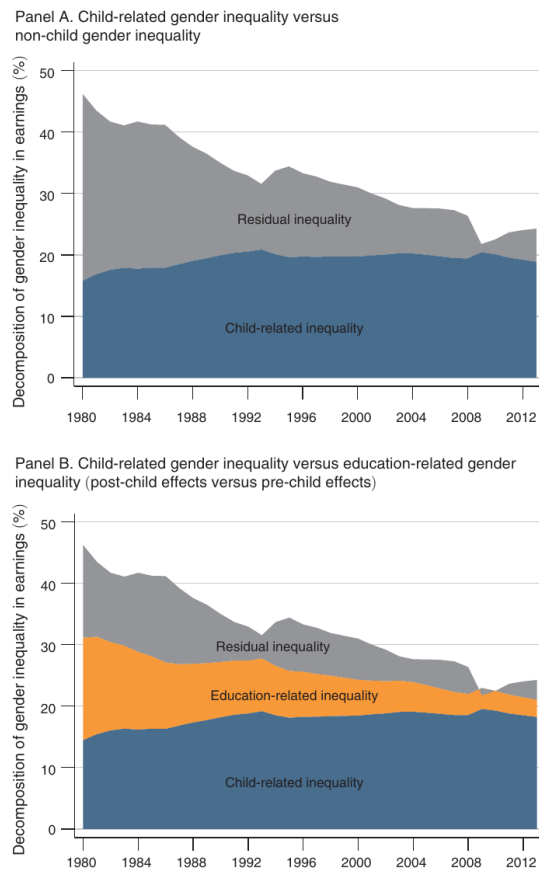
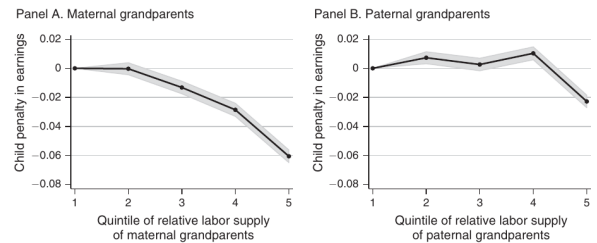


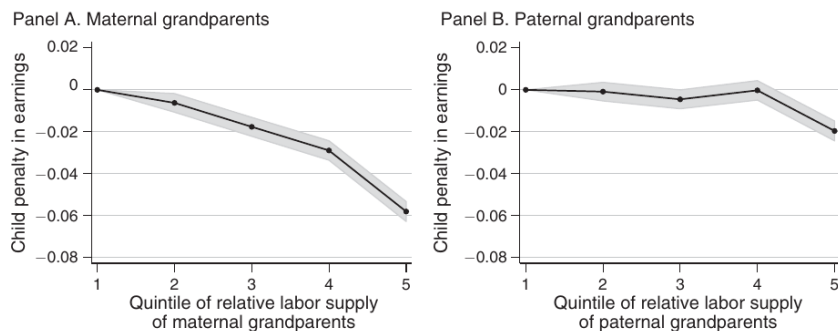
Figure 6: Intergenerational Transmission of Child Penalties without Controls



Read. The decomposition visual shows the central historical result: child-related inequality becomes the dominant component of the gender earnings gap over time. The intergenerational visual without controls shows that women from more modern maternal-grandparent households have smaller child penalties, while the paternal-grandparent relationship is weak.

6.4 Figure 7: Intergenerational Transmission of Child Penalties with Controls

Figure 7: Intergenerational Transmission of Child Penalties with Controls



Read. Adding controls for grandparent characteristics does not eliminate the maternal-grandparent pattern. This supports the interpretation that the relationship is not simply driven by education, wealth, region, or cohort differences across families.

7 Short summary

- **Method.** Full-population Danish administrative data plus event studies around the birth of the first child.
- **Main empirical result.** Women’s labor-market outcomes fall sharply after childbirth, while men’s outcomes are almost unchanged. The long-run earnings penalty is about 20 percent.
- **Mechanism.** The penalty reflects participation, hours, wages, occupational rank, management opportunities, and movement toward family-friendly jobs and firms.
- **Decomposition result.** Child penalties explain a rising share of earnings inequality: roughly 40 percent in 1980 and 80 percent by 2013.
- **Intergenerational result.** Child penalties are transmitted through maternal family background. More traditional maternal-grandparent households predict larger penalties for daughters who become mothers.
- **Conceptual contribution.** The paper reframes gender inequality as a parenthood problem. Remaining inequality is not just about unequal pay for equal work; it is also about unequal allocation of work after children.

8 Conclusion

Core conclusion. Children create large, persistent, and strongly gendered labor-market penalties. In Denmark, these penalties explain most of the remaining gender earnings inequality.

Economic intuition. Before children, men and women are on similar labor-market paths. After children, women adjust on multiple margins: they work less, participate less, earn lower wages,

move away from management trajectories, and sort into more family-friendly jobs. Men do not make comparable adjustments.

Takeaway. Gender equality in education and formal labor-market access is not enough to eliminate earnings inequality. As long as child-care responsibilities and family-career tradeoffs remain gendered, parenthood will continue to reproduce gender inequality in the labor market.